batch: b2 Roll No.: 107

Experiment No. 02

Grade: AA / AB / BB / BC / CC /

TITLE: Shell Programming

AIM: To study the shell script and write the program using shell.

Expected Outcome of Experiment:

CO 1. To introduce basic concepts and functions of operating systems.

Books/ Journals/ Websites referred:

1. **Silberschatz A., Galvin P., Gagne G. "Operating Systems Principles", Willey Eight edition.**
2. **William Stallings "Operating Systems" Person, Seventh Edition Edition.**
3. **Sumitabha Das " UNIX Concepts & Applications", McGraw Hill Second Edition.**

Pre Lab/ Prior Concepts:

The shell provides you with an interface to the UNIX system. It gathers input from you and executes programs based on that input. When a program finishes executing, it displays that program's output.

Shell Scripts
The basic concept of a shell script is a list of commands, which are listed in the order of execution. A good shell script will have comments, preceded by a pound sign, #, describing the steps.

Steps to create a Shell Script:
Create a file using any text editor say vi, gedit, nano etc
1. $ vi filename
2. Insert the script/ commands in file and save the file to execute the file we need to give execute permission to the file
3. $ chmod 775 filename
4. Now execute the above file using any of following methods:
$ sh filename OR $ ./filename

NOTE: Before adding anything to your script, you need to alert the system that a shell script is being started. This is done using the shebang construct. For example –
#!/bin/sh.

Description of the application to be implemented:

1. Write a shell Script that accepts two file names as command line arguments and compare two file contents and check whether contents are same or not. If they are same, then delete second file.

*Duplicate.sh
~/Ashwera

```
echo "Enter the names of the files"
read f1
read f2

if cmp -s "$f1" "$f2"; then
        rm -- "$f1"
else
        echo "No duplicates found"
fi
```

```
kjsce@kjsce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Ashwera$ touch file1.txt
kjsce@kjsce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Ashwera$ nano file1.txt
kjsce@kjsce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Ashwera$ cat file1.txt
ashwera
kjsce@kjsce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Ashwera$ cat file2.txt
ashwera
kjsce@kjsce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Ashwera$ cat file.txt
riya
kjsce@kjsce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Ashwera$ sh Duplicate.sh
Enter the names of the files
file1.txt
file.txt
No duplicates found
kjsce@kjsce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Ashwera$ sh Duplicate.sh
Enter the names of the files
file1.txt
file2.txt
kjsce@kjsce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Ashwera$ ls
Duplicate.sh  file2.txt  file.txt
kjsce@kjsce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Ashwera$
```

2. 2. Write a shell script that accepts integer and find the factorial of number.

```
Enter the number
4
24
kjsce@kjsce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Ashwera$ sh Factorial.sh
Enter the number
5
120
kjsce@kjsce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Ashwera$
```

```
echo "Enter the number"
read x
sum=1
while [ "$x" -gt 0 ]
do
        sum=$((sum * x))
        x=$((x - 1))
done
echo $sum
```

3. Write a shell script for adding users

```
Sheldon:1j23jnfajhda
Leonard:adbsbdajdakdb
Penny:adsasjdbakdbaj
Raj:92431hadsjkbadakd
Howard:ajdbjasdba
```

```
while IFS=: read -r username password
do
    echo "Creating user: $username"
    sudo useradd "$username"
    echo "$username:$password" | sudo chpasswd
done < names.txt
```

```
purva:x:1015:1015::/home/purva:/bin/sh
rahul:x:1016:1016::/home/rahul:/bin/sh
neha:x:1017:1017::/home/neha:/bin/sh
Ashwera:x:1018:1018::/home/Ashwera:/bin/sh
sanvi:x:1019:1019::/home/sanvi:/bin/sh
Molaika:x:1020:1020::/home/Molaika:/bin/sh
Sheldon:x:1021:1021::/home/Sheldon:/bin/sh
Leonard:x:1022:1022::/home/Leonard:/bin/sh
Penny:x:1023:1023::/home/Penny:/bin/sh
Raj:x:1024:1024::/home/Raj:/bin/sh
Howard:x:1025:1025::/home/Howard:/bin/sh
```

4. Write a shell script for counting no of processes running on system

```
kjsce@kjsce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Ashwera$ gedit Processes.sh
kjsce@kjsce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Ashwera$ sh Processes.sh
Number of running processes: 477
kjsce@kjsce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Ashwera$
```

```
count=$(ps -e | wc -l)
count=$((count - 1))

echo "Number of running processes: $count"
```

Modification-

Write a shell script for counting the no of logged in users.

```
count=$(who | wc -l)

echo "Number of logged in users: $count"
```

```
kjsce@kjsce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Ashwera$ gedit Processes.
kjsce@kjsce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Ashwera$ sh Processes.sh
Number of logged in users: 2
kjsce@kjsce-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Ashwera$
```

Conclusion :

This shell script successfully counts logged-in users. It uses standard UNIX commands. The output is accurate and simple.

Post Lab Descriptive Questions

1) What are the different types of commonly used shells on a linux system?

Commonly used Linux shells:

- Bourne Shell (sh)
- Bash (bash)
- C Shell (csh)
- TC Shell (tcsh)
- Korn Shell (ksh)
- Z Shell (zsh)

2) How do you find out the current shell that you are working on?

We find out the current shell we are working on by using this command:
echo $SHELL
The ps command can also be used.

3) List the advantages and disadvantages of shell scripting.

Advantages:

- Automates repetitive tasks
- Saves time and effort
- Easy to write and modify
- Uses existing UNIX commands

Disadvantages:

- Slower than compiled languages
- Limited for complex programs
- Harder to debug large scripts

- Portability issues across shells

Date: 28/01/2026 **Signature of faculty in-charge**